CORRELATION

The correlation is a statistical tool which studies the relationship between two variables and correlation analysis involves various methods and techniques used for studying and measuring the extent of the relationship between the two variables.

Two variables are said to be correlated if the change in one variable results in a corresponding change in the other variable.

Types of Correlation

(a) POSITIVE AND NEGATIVE CORRELATION

If the values of the two variables deviate in the same direction i.e., if the increase in the values of one variable results, on an average, in a corresponding increase in the values of the other variable or if a decrease in the values of one variable results on an average in a corresponding decrease in the values of the other variable, correlation is said to be **positive**.

Positive Corr ↑↑ or ↓↓

Some examples of series of positive correlation are:

i) Heights and weights
ii) The family income and expenditure on luxury items
iii) Annual mileage and annual maintenance costs
iv) The age and value of a computer
v) The day temperature and sale of soft drinks
vi) The amount of rainfall and yield of a crop (up to a point)
vii) Price and supply of a commodity and so on

On the other hand, correlation is said to be negative or inverse if the variables deviate in opposite direction i.e., if the increase (decrease) in the values of one variable results, on an average, in a corresponding decrease (increase) in the values of the other variable.

Some examples of negative correlation are the series related to:

i) Price and demand of a commodity
ii) The age and second-hand value of cars
iii) Height above sea level and temperature

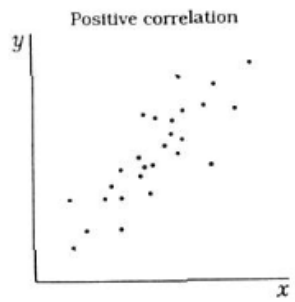
iv) Volume and pressure of a perfect gas
v) The day temperature and sale of woolen garments, and so on.

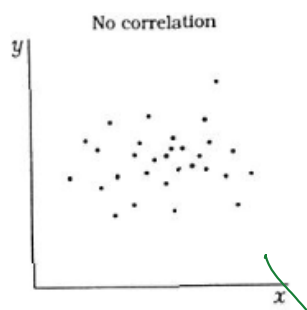
Methods of studying correlation

We shall confine our discussion to the methods of ascertaining only **linear relation between two variable**s or series.
The commonly used methods for studying the correlation between two variables are:

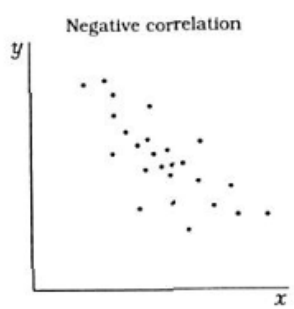
i) Scatter diagrams method
ii) Karl Pearson's coefficient of correlation (covariance method)
iii) Rank method

The method of scatter diagram only tells us about the nature of the relationship whether it +ve or -ve and whether it is high or low. It does not provide us an exact measure of the extent of the relationship between the two variables.





zero Corr



Karl Pearson's Coefficient of Correlation (Covariance Method)

A mathematical method for measuring the intensity or the magnitude of linear relationship between two variable series was suggested by Karl Pearson (1867-1936) a British Biometrician and Statistician and is by far the most widely used method in

relationship between two variable series was suggested by Karl Pearson (1867-1936) a British Biometrician and Statistician and is by far the most widely used method in practice.

$d_x = X - \bar{X}$; $d_y = y - \bar{y}$

Symbolically,

$$r = \frac{Cov,(x,y)}{\sigma_x \sigma_y} = \frac{\sum d_x d_y}{\sqrt{d^2{}_x d^2{}_y}}$$

$r = 0 \Rightarrow$ no Corr.

$r = +1 \Rightarrow$ perfect +ve Corr

Where d_x and d_y denote the deviations of x and y values from their arithmetic means $\bar{x}$ *and* $\bar{y}$ *respectively*.

$r = -1 \Rightarrow$ perfect Inverse Corr.

PROPERTIES OF CORRELATION COEFFICIENT

Property I: Limits for Correlation Coefficient

Pearsonian correlation coefficient cannot exceed 1 numerically. In other words it lies between -1 and +1.

Symbolically: $-1 \leq r \leq +1$

Remarks:

1. This theorem provides us a check on our calculations. If in any problem, the obtained value of r lies outside limits ± 1, this implies there is some mistake in our calculations.
2. Correlation coefficient is independent of the change of origin and scale. Mathematically, if X and Y are the given variables and they are transformed to the new variables U and V by the change of origin and scale viz.,

$$u = \frac{x - A}{h} \quad and \quad v = \frac{y - B}{k}$$

We have: $r_{xy} = \frac{\sum(u-\bar{u})(v-\bar{v})}{\sqrt{\sum(u-\bar{u})^2 \sum(v-\bar{v})^2}} = \frac{n\sum uv - (\sum u)(\sum v)}{\sqrt{\sum(u-\bar{u})^2 \sum(v-\bar{v})^2}} = r_{uv}$

$$\therefore r_{uv} = \frac{n\sum uv - (\sum u)(\sum v)}{\sqrt{[n\sum u^2 - n(\sum u)^2][n\sum v^2 - n(\sum v)^2]}}$$

Example

Calculate the coefficient of correlation for the ages of husband and wife:

Age of husband (x)	23	27	28	29	30	31	33	35	36	39
Age of wife (y)	18	22	23	24	25	26	28	29	30	32

Use the Covariance Method.

Solution

Calculations for Correlation Coefficient

x	y	u=x-30	v=y-25	u^2	v^2	uv
22	18	-9	-7	81	49	63
27	22	-4	-3	16	9	12
28	23	-3	-2	9	4	6
29	24	-2	-1	4	1	2
30	25	-1	0	1	0	0
31	26	0	1	0	1	0
33	28	2	3	4	9	6
35	29	4	4	16	16	16
36	30	5	5	25	25	25
39	32	8	7	64	49	55
$\sum x=310$	$\sum y=257$	$\sum u=0$	$\sum v=7$	$\sum u^2=220$	$\sum v^2=163$	$\sum uv=179$

$$r_{uv} = \frac{n\sum uv - (\sum u)(\sum v)}{\sqrt{[n\sum u^2 - (\sum u)^2][n\sum v^2 - (\sum v)^2]}}$$

$$= \frac{10 \times 179 - 0 \times 7}{\sqrt{[10 \times 220 - (0)^2][10 \times 163 - (7)^2]}}$$

$$= \frac{1790 - 7}{\sqrt{(2020 - 0)(1630 - 49)}} = \frac{1783}{\sqrt{2020 \times 1581}} = 0.9977224017$$

Since Karl Pearson's correlation coefficient r is independent of change of origin, we get

$r_{xy} = r_{uv} = 0.9977$ There is a very close correlation between the two valriables

Example

The co-ordinates given below refer to an experiment to verify Newton's law of cooling over a limited range of values. Determine the value, degree and nature of the coefficient of correlation.

Time (min)	4	8	10	12	16	22
Temperature (^{0}C)	46	34	30	26	24	20

Solution

x	y	u=(x-12)/2	v=(y-30)/2	u^2	v^2	uv
4	46	-4	8	16	64	-32
8	34	-2	2	4	4	-4
10	30	-1	0	1	0	0
12	26	0	-2	0	4	0
16	24	2	-3	4	9	-6
22	20	5	-5	25	25	-25
$\sum$x=72	$\sum$y=180	$\sum$u=0	$\sum$v=0	$\sum u^2$=50	$\sum v^2$=106	$\sum$uv=-67

We have: $r_{xy} = \frac{\sum(u-\bar{u})(v-\bar{v})}{\sqrt{\sum(u-\bar{u})^2\sum(v-\bar{v})^2}} = \frac{-67}{\sqrt{50\times106}} = -0.9203157785$

$\therefore value = -0.920$

Degree=good

Nature=inverse

HOMEWORK

Find the Karl Pearson's correlation coefficient between sales (in '000 of units) and expenses (in '00,000 of Ksh) of the following ten firms

Firm	1	2	3	4	5	6	7	8	9	10
Sale	50	50	55	60	65	65	65	60	60	50
Expenses	11	13	14	16	16	15	15	14	13	13

Also mention what does your calculated value of r indicate

Rank Correlation Method

Sometimes we come across statistical series in which the variables under consideration are not capable of quantitative measurement but can be arranged in a serial order.

This happens when we are dealing with qualitative characteristics (attributes) such as honesty, beauty, character, morality, etc., which cannot be measured quantitatively but can be arranged serially.

Is such situations Karl Pearson's coefficient of correlation cannot be used as such.

Charles Edward Spearman, a British Psychologist, developed a formular in 1904 which consists in obtaining the correlation coefficient between the ranks of n individuals in the **two** attributes under study.
Spearman's rank correlation coefficient, usually denoted by β (Rho)is given by

$$\beta = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

Where d is the difference between the pairs of ranks of the same individual in the two characteristics and n is the number of pairs.

NB/
Spearman's rank correlation coefficient lies between -1 and +1

$$-1 \leq \beta \leq +1$$

Spearman's rank correlation coefficient formula can also be used when the actual data, but not the ranks relating to two variable, are given.
In such case we shall have to convert the data into ranks.
The highest (smallest) observation is given the rank 1.
The next highest (smallest) observation is given rank 2 and so on.
It is immaterial in which way (ascending or descending) the ranks are assigned.
However, the same approach should be followed for all the variable under consideration.

Example

Calculate Spearman's rank correlation coefficient between advertisement cost (in '00,000 Ksh) and sales (in '000 of units) from the following data:

Advert (x)	39	65	62	90	82	75	25	98	36	78
Sales (y)	47	53	58	86	62	68	60	91	51	84

Solution

X	Y	Rank of X (x)	Rank of Y (y)	d=x-y	d^2
39	47	8	10	-2	4
65	53	6	8	-2	4
62	58	7	7	0	0
90	86	2	2	0	0
82	62	3	5	-2	4
75	68	5	4	1	1

25	60	10	6	4	16
98	91	1	1	0	0
36	51	9	9	0	0
78	84	4	3	1	1
					$\sum d^2=30$

$$\text{þ} = 1 - \frac{6\times30}{10(100-1)} = 0.82$$

REPEATED RANKS

In case of attributes if there is a tie, then common ranks are assigned to the repeated items.

These common ranks are the arithmetic mean of the ranks which these items would have got if they were different from each other and the next item will get the rank next to the rank used in computing the common rank.

For example, suppose an item is repeated at rank 4.

Then the common rank to be assigned to each item is (4+5)/2, i.e., 4.5

The next item will be assigned rank 6.

If the item is repeated thrice, at rank 7, then the common rank to be assigned to each item will be (7+8+9)/3, i.e., 8

The next rank to be assigned will be rank 10.

In the Spearman's rank formula add the factor $\frac{m(m^2-1)}{12}$ to $\sum d^2$ where m is the number of times an item is repeated.

This correction factor is to be added for each repeated value in both the series.

Example

A psychologist wanted to compare two methods A and B of teaching. He selected a random sample of 22 students. He grouped them into 11 pairs so that the students in a pair have approximately equal scores on an intelligence test. In each pair one student was taught by method A and the other by method B and examined after the course. The marks obtained by them are tabulated below:

Pair	1	2	3	4	5	6	7	8	9	10	11
A	24	29	19	14	30	19	27	30	20	28	11
B	37	35	16	26	26	27	19	20	16	11	21

Determine the Spearman's rank correlation coefficient

Solution:

1.5 1.5 3 4 5 6 7 8.5 8.5 10 11

30 30 29 28 27 24 20 19 19 14 11

Calculations of Rank Correlation Coefficient

X	Y	Rank of X (x)	Rank of Y (y)	d=x-y	d^2
24	37	6	1	5	25
29	35	3	2	1	1
19	16	8.5	9.5	-1	1
14	26	10	4	6	36
30	23	1.5	5	-3.5	12.25
19	27	8.5	3	5.5	30.25
27	19	5	8	-3	9
30	20	1.5	7	-5.5	30.25
20	16	7	9.5	-2.5	6.25
28	11	4	11	-7	49
11	21	11	6	5	25
					$\sum d^2=225$

Hence we see that in the x-series the items 19 and 30 are repeated, each occurring twice and in the y-series the item 16 is repeated.
Thus, in each of the three case m=2.

Hence, on applying the correction factor $\frac{m(m^2-1)}{12}$ for each repeated item, we get

$$\text{þ} = 1 - \frac{6[\sum d^2 + \frac{2(4-1)}{12} + \frac{2(4-1)}{12} + \frac{2(4-1)}{12}]}{11(121-1)} = 1 - \frac{6 \times 226.5}{11 \times 120}$$

þ=1-1.0225=-0.0225

EXAMPLE

The ranks of the same 15 students in two subjects A and B are given below; the two numbers within brackets denoting the ranks of the same students in A and B respectively.
(1,10), (2,7), (3,2), (4,6), (5,4), (6,8), (7,3), 8,1, (9,11), (10,15), (11,9),(12,5), (13,14), (14,12), (15,13)
Use the Spearman's formular to find the rank correlation coefficient.

SOLUTION

CALCULATION OF SPEARMAN'S CORRELATION COEFFICIENT

Rank in A (x)	Rank in B (y)	d=x-y	d^2
1	10	-9	81
2	7	-5	25
3	2	1	1
4	6	-2	4
5	4	1	1
6	8	-2	4
7	3	4	16
8	1	7	49
9	11	-2	4
10	15	-5	25
11	9	2	4
12	5	7	49
13	14	-1	1
14	12	2	4
15	13	2	4
			$\sum d=272$

Spearman's rank correlation coefficient þ is given by

$$þ= 1 - \frac{6\sum d^2}{n(n^2-1)} = 1 - \frac{6\times 272}{15(225-1)} = 0.51$$

HOMEWORK
Quotations of index number of security prices of a certain joint stock company and of

prices of preference shares and debentures are given below:

Preference share price (x)	73.2	85.8	78.9	77.2	81.2	83.8
Debenture price (y)	97.8	99.2	98.8	98.3	96.7	97.1

Use the method of rank correlation to determine the relationship between prices and debenture prices.

ASSGT

Ten competitors in a beauty contest are ranked by three judges in the following order:

1st Judge	1	6	5	10	3	2	4	9	7	7
2nd Judge	3	5	8	4	7	10	2	1	6	9
3nd Judge	6	4	9	8	1	2	3	10	5	7

Use the Spearman's rank correlation coefficient to determine which pair of judges has nearest approach to common taste in beauty.